Introduction

Background: Vehicle fires represent a critical issue across all types of cars.

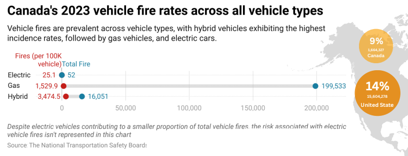


Figure 1: Vehicle Fire Rates Across All Vehicle Types in Canada, 2023

- **68 fires per 100,000 cars** when taking all fuel types into account.
- Until July 26 in 2022, **400 motor vehicle safety recalls** have been issued in Canada and among those, **92** were to address the risk of fire, affecting **650,937** vehicles

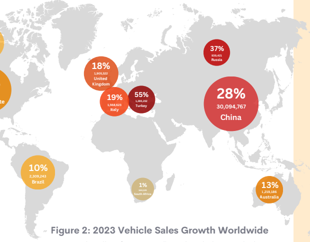


Figure 2: 2023 Vehicle Sales Growth Worldwide

Data Source: https://www.factorywarrantylist.com/car-sales-by-country.html

Background: Vehicle fires pose increased hazards in electric vehicles (EVs)

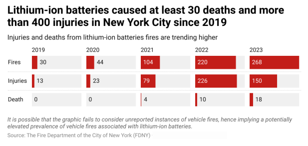


Figure 3: Vehicle Fires in New York City caused by lithium-ion batteries

- Electric Vehicles (EVs) have about **1,000** times more cells than an e-bike

  Increasing number of cells with a higher energy has a greater risk during failure

- An average of **93** auto fires yearly in Canada linked to safety defects

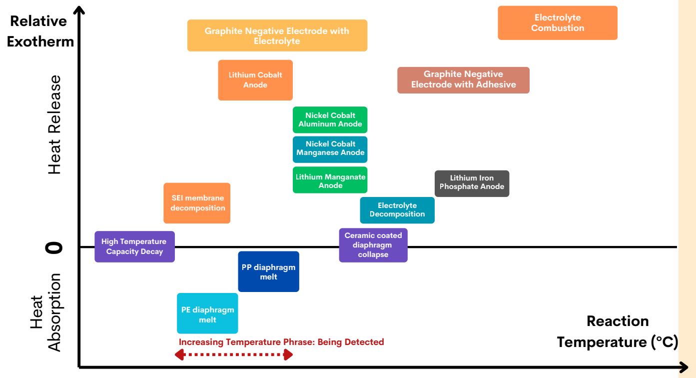


Figure 4: Chemical Reactions During Thermal Runaway in Lithium-ion Batteries

Challenge: Vehicle Door Release System Malfunctions are Real

Nov 5th, 2021
Parking lot, Florida, US
Use a fire extinguisher to break the passenger window and unlock SUV doors

July 4th, 2022
Queen Elizabeth Highway, Mississauga, Ontario, Canada (shown at right)
Break the window using hammer and pulled the driver out

July 26th, 2023
Along a highway in California, US
Pull the driver away from the burning car right before the fire built back up again

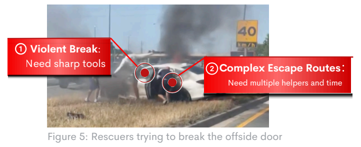


Figure 5: Rescuers trying to break the offside door

Figure 6: Fire built back up again in 15 seconds

Project Goals and Objectives:

Goals:

1) Develop an automatic emergency **door release mechanism** to enhance survivability in vehicle fires using a combination of existing equipment and custom components.
2) Demonstrate the mechanism's reliability by testing its **activation time, performance consistency** and **adaptability** under different circumstances.

Objectives:

1) Develop an automatic emergency door release mechanism to:
   a. Operate **independently** of the vehicle central control system.
   b. Activate upon **early detection of thermal runaway** in lithium-ion batteries.
2) Validate the mechanism's reliability to:
   a. Determine the **response time** over several experimental replications.
   b. Determine the impact of **temperature variations** on response time.
   c. Determine the **minimum air pressure** the mechanism needs to operate.
   d. Evaluate the overall **robustness** of the activation mechanism by analyzing its response time and success rate.

Methodology

Methodology Step 1): Prototype of the system

Objective: Operate independently of the vehicle central control system

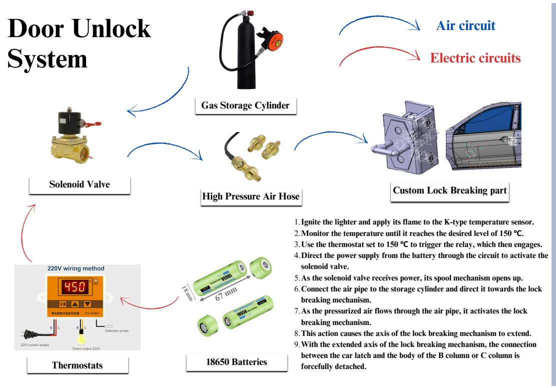


Figure 6: Configuration and principles of the door release mechanism

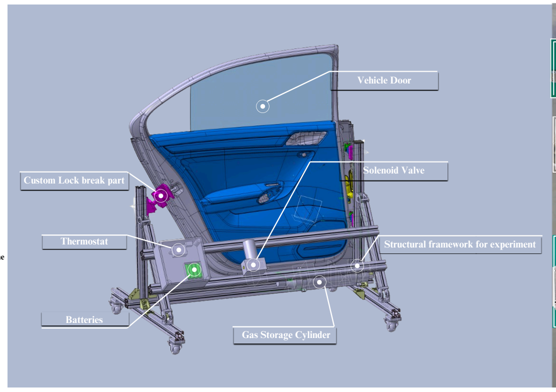


Figure 7: Model of the door release mechanism

Methodology Step 3): Experiment Setup

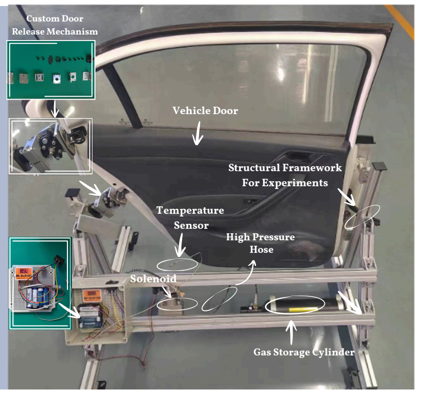


Figure 8: Experimental Setup of the door release mechanism

Methodology Step 2): Design of the Custom Lock Breaking Component

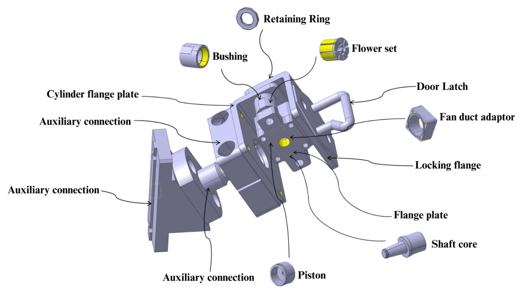


Figure 9: Model of the Custom Lock Breaking Component

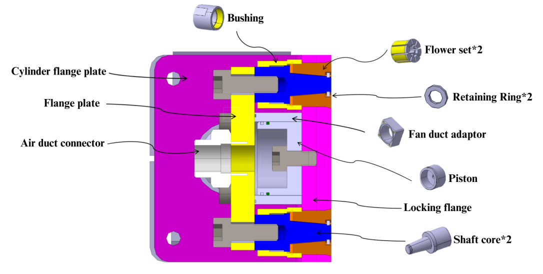


Figure 10: Layout Schematic Graph for the Custom Lock Breaking Component

Mechanism for the Custom Door Release Component

1. Compressed air from the gas pipe connector into the inner cavity of the cylinder
2. The piston is fixed → the cylinder is driven by high-pressure gas → piston displacement with the cylinder
3. The cylinder pushes the cylinder flange action → the cylinder flange drives the axle to pull out of the flower sleeve
4. The axle sleeve and the flower sleeve are connected through threads; the axle sleeve and the axle are correlated through the stop limit
5. When the axle to the back of the time-ordered action, the axle and the flower sleeve clearance between the flower sleeve to make the flower sleeve have the Elastic variable
6. Then the axle center pulls the axle sleeve backward, driving the flower sleeve out of the locking flange plate
7. The locking flange has no connection point, the door is opened.
8. Note:the role of the ring is to prevent the door from opening by mistake and design an error-proof mechanism.

Results

Result 1): Robustness Analysis: under Different Temperature and Uniform Air Pressure Conditions

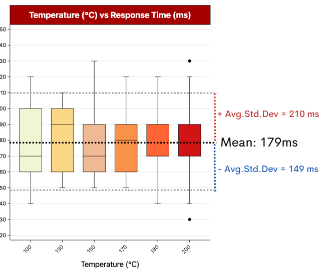
The robustness of the door release mechanism is evaluated by examining the distribution of response times, reflecting variations in the duration for the door to fully release. This analysis encompasses thirty tests divided into six groups, each with the temperature activation set at 100°C, 130°C, 150°C, 170°C, 180°C, and 200°C, respectively. The air pressure is consistently controlled at 15 MPa across all tests.

Temperature variation (°C) vs Response time (ms)

In each temperature scenario (100°C, 130°C, 150°C, 170°C, 180°C, 200°C), five tests were performed, revealing consistent response times across the temperature range, indicative of stability in the system's performance regardless of temperature variation

	100°C	130°C	150°C	170°C	180°C	200°C
A	170	160	170	150	220	170
B	220	150	190	180	140	130
C	140	210	150	160	190	170
D	200	190	230	190	170	230
E	160	200	160	220	170	190

In the third test, selective substitution of parts in the custom metal component is employed to reduce potential errors resulting from wear and tear on the component.



Statistical Analysis: Anova Test (α=0.05)

Temperature(°C)	N	Mean	Std.Dev	Std.Error
100	5	178	32	14
130	5	182	26	12
150	5	180	32	14
170	5	180	27	12
180	5	178	29	13
200	5	178	36	16

Source	*Degree of Freedom*	*Sum of Squares*	*Mean Square*
Between Groups	5	67	13
Within Groups	5	22,520	938
Total	29	22,587	

F = Mean Square Between Group/Mean Square Within Group
= 67 / 22520 ≈ 0.014

P = Right Tail of (DF between groups, DF within groups)
≈ 0.9992 > 0.05

∵ Since p-value > α, H_0 (null hypothesis) is accepted.
The averages of all groups assumed to be equal.

Conclusion:
Variations in temperature have no discernible impact on response time, nor was any expected.

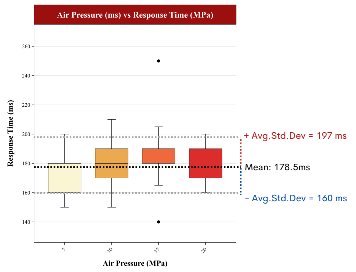
Result 2): Robustness Analysis: under Uniform Temperature and Different Air Pressure Conditions

The robustness of the door release mechanism is evaluated by examining the distribution of response times, reflecting variations in the duration for the door to fully release. This analysis encompasses eighteen tests divided into six groups, each with the air pressure in the cylinder set at 3MPa, 5MPa, 10MPa, 15MPa and 20 MPa, respectively. The temperature activation point is consistently controlled at 150°C across all tests.

Pressure Variation (MPa) vs Response time (ms)

Pressure (MPa)	1st Response time(ms)	2nd Response time(ms)	3rd Response time(ms)	4th Response time(ms)	5th Response time(ms)
<5	N/A	N/A	N/A	N/A	N/A
5	180	200	150	160	180
10	210	190	170	170	180
15	190	190	140	190	180
20	170	190	170	200	160

In the third test, selective substitution of parts in the custom metal component is employed to reduce potential errors resulting from wear and tear on the component.



The failure point of the mechanism occurs when the air pressure within the cylinder drops to 3MPa. This indicates a critical threshold between 3MPa and 5MPa, below which the system fails to function effectively.

Statistical Analysis: Anova Test (α=0.05)

Air Pressure (Mpa)	N	Mean	Std.Dev	Std.Error
5	5	174	19	9
10	5	184	17	7
15	5	178	22	10
20	5	178	16	7

Source	Degrees of Freedom	Sum of Squares	Mean Square
Between Groups	3	255	85
Within Groups	16	5,600	350
Total	19	5,855	

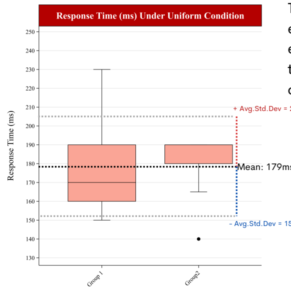
F = Mean Square Between Group/Mean Square Within Group
= 85 / 350 ≈ 0.024

P = Right Tail of (DF between groups, DF within groups)
≈ 0.8652 > 0.05

∵ Since p-value > α, H (null hypothesis) is accepted.
The averages of all groups assumed to be equal.

Conclusion:
Variations in air pressure level have no discernible impact on response time above a critical point between 3-5 MPa, nor any was expected.

Result 3): Replicability Test: Consistency under Uniform Temperature and Air Pressure across Different Days



The robustness of the door release mechanism is evaluated by examining the consistency of response times. This analysis encompasses ten tests divided into two groups, with the temperature activation set at 150°C and air pressure being consistently controlled at 15 MPa across all tests.

Comparative Analysis of Response Time (ms) at 150°C Activation and 15 MPa Air Pressure

Ten tests conducted under identical conditions are partitioned into two distinct groups and executed on separate days, highlighting the consistency and stability of the mechanism through repeated trials under uniform conditions.

	Group 1	Group 2
A	170	190
B	190	190
C	150	140
D	230	190
E	160	180

Statistical Analysis: T Test (α=0.05)

Groups	N	Mean	Std.Dev	Std.Error
Group 1	5	180	32	14
Group 2	5	178	22	10

Source	Degrees of Freedom	Sum of Squares	Mean Square
Between Groups	1	10	10
Within Groups	8	5,880	735
Total	9	5,890	

F = Mean Square Between Group/Mean Square Within Group
= 10 / 735 ≈ 0.014

P = Right Tail of (DF between groups, DF within groups)
≈ 0.91 > 0.05

∵ Since p-value > α, H (null hypothesis) is accepted.
The averages of all groups assumed to be equal.

Conclusion:
Response time remains consistent across different days under uniform conditions, with no significant impact observed on response time variability. This outcome aligns with the anticipated stability of the mechanism

Discussion:

Uncertainties and Limitations:

The processing limitations of LoggerPro and signal transmission uncertainties resulted in a **precision limitation** to two significant figures, with an associated **uncertainty of 1 millisecond**. Despite this, the precision remains acceptable given the negligible impact of this level of uncertainty in the context of a vehicle fire.

While I tested two independent variables (i.e., temperature and air pressure), **other factors** such as humidity and smoke were not accounted for. This omission may introduce variability in the results. A future study could look at these effects.

Although the comparison between battery combustion time and escape time is the main consideration in this project, **the use of real batteries was prohibited** due to safety concerns, limiting the direct assessment of this door release mechanism.

The custom door release mechanism needed parts replaced after wearing out in the experiment, indicating it is best used as a **one-time device**. This is because, in a vehicle fire, the mechanism is designed to work once and then be replaced to maintain its functionality, similar to other safety equipment such as an air bag.

Conclusion:

The project goals were met:

1) The system works **independently** from the vehicle's central control system, significantly improving its reliability in various situations.

2) The system activates upon **early detection of thermal runaway** in lithium-ion batteries.

3) Three experiments showed that changes in temperature and air pressure **don't affect the response time's consistency**, even under repeated trials with the same conditions.

4) An average response time of **≤200ms**, akin to an eye blink, is detected in experiments. This is crucial in vehicle fires because it enables near-instantaneous door release for rapid evacuation. This rapid activation is vital for enhancing survival probabilities and minimizing injury risks during the time-sensitive scenario of a vehicle fire.

Future Work:

1) Integrating **additional sensors**, including smoke and temperature detectors, across various vehicle components could refine the accuracy of early thermal runaway detection and other pre-combustion indicators.

2) Future work should explore **the inclusion of batteries** to assess the interval between battery ignition and viable escape opportunities, considering door release timings. Such investigations are expected to yield insights into actual escape durations.

3) Subsequent research should examine a wider range of scenarios to design an autonomous door release system capable of **facilitating unassisted escapes from vehicle fires**, even in scenarios where occupants are incapacitated, minimizing risk to both occupants and potential rescuers.

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Thermostats

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